

MUHAMMAD HAMMAD ASHER

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SUMMARY

Software Engineering student at Bahria University Karachi with a strong foundation in full-stack development, MLOps, and machine learning. Delivered end-to-end projects spanning production-grade MLOps pipelines, real-time ETL systems, and role-based web applications. Seeking an internship in software engineering or AI/ML engineering.

EDUCATION

Bahria University, Karachi <i>Bachelor of Software Engineering</i> Relevant Coursework: Data Structures & Algorithms, Operating Systems, DBMS, Software Design & Architecture, OOP, Design & Analysis of Algorithms	2024 – Present CGPA: 3.63 / 4.00
Saylani Mass IT Training (SMIT) <i>Course in Artificial Intelligence & Data Science</i>	July 2025 – Present

ACHIEVEMENTS & COMPETITIONS

1st Place — Agentic AI Track, Techathon 2.0 • Designed and presented a multi-agent AI pipeline competing against university teams across Pakistan; won first place in the Agentic AI category.	2026
Runner-Up — The Heist, Procom 2025 (FAST NUCES) • Placed 2nd in a nation-level competitive problem-solving event hosted by one of Pakistan's top engineering universities.	2025

TECHNICAL SKILLS

Languages: Python, Java, C++, C#, JavaScript, SQL
Frontend: HTML5, CSS3, Bootstrap, JavaScript (ES6+), DOM Manipulation
Backend & APIs: Flask, FastAPI, RESTful API Design
Databases: MySQL, SQL Server, XML-based Storage
AI / ML & MLOps: scikit-learn, TensorFlow, Keras, CNNs, RNNs, MLflow, DVC, Evidently AI, Great Expectations, PySpark, Audio/Sound Classification
Data Engineering: ETL Pipelines, pandas, NumPy, Matplotlib, Plotly, Streamlit
Deployment: Docker, Kubernetes (basics), Streamlit Cloud, Render, AWS (basics), Hugging Face, DagsHub
Tools & Platforms: Git, GitHub, GitHub Actions (CI/CD), VS Code, Windows Forms (.NET), IntelliJ
Concepts: OOP, Data Structures & Algorithms, Design Patterns (MVC, Factory, Observer, Singleton, Decorator), SDLC, Agile

PROJECTS

Bank Marketing Term Deposit Prediction — MLOps Pipeline Live App <i>Python · scikit-learn · MLflow · DVC · FastAPI · PySpark · Docker · Streamlit</i> • Engineered a production-grade MLOps pipeline on the UCI Bank Marketing dataset (45,211 rows) covering data ingestion, validation via Great Expectations, SMOTE-based class balancing, feature engineering, and model training orchestrated with DVC. • Integrated MLflow for experiment tracking and hyperparameter tuning; implemented probability calibration via Platt scaling with a CI/CD model gate blocking deployment if F1 < 0.60 or AUC < 0.75. • Containerised the FastAPI inference endpoint and Streamlit dashboard with Docker; configured GitHub Actions for automated testing, staging, and production deployment with Evidently AI drift detection (PSI > 0.2) triggering scheduled weekly retraining.	
Weather & Finance ETL Analytics Dashboard Live App <i>Python · Streamlit · pandas · Plotly · REST APIs</i> • Architected a real-time ETL pipeline ingesting and normalising data from OpenWeatherMap and Alpha Vantage APIs, processing metrics for 15,000+ global entities.	

- Built interactive Plotly dashboards surfacing financial indicators (Moving Averages, Volume Analysis) and atmospheric metrics; deployed on Streamlit Cloud with zero-downtime access.
- Applied pandas transformation layers for data cleaning, type coercion, and aggregation, reducing raw API payload size by retaining only relevant feature columns downstream.

Personal Portfolio Website [Live Site](#)

HTML5 · CSS3 · JavaScript ES6 · Glassmorphism UI

- Built a fully responsive single-page developer portfolio with dark/light theme toggle, glassmorphism card design, scroll-triggered animations via IntersectionObserver, and project category filtering.
- Implemented a time-aware hero section with an animated code-window mockup, interactive education timeline, staggered skill badge animations, and a client-side validated contact form, all in vanilla HTML, CSS, and JavaScript with zero dependencies.

Hospital Management System

Python · Flask · Microsoft SQL Server · Jinja2 · HTML5 · CSS3 · Flask-Login

- Built a role-based web application (Administrator, Doctor, Nurse, Billing Staff) using Flask and MVC architecture, covering patient registration, appointment scheduling with conflict detection, pharmacy inventory with low-stock alerts, and itemised invoice generation backed by Microsoft SQL Server.
- Implemented secure authentication with Flask-Login and bcrypt, enforced RBAC scoping across all modules, and delivered an admin dashboard with real-time stats (active patients, today's appointments, revenue, inventory status) alongside exportable reports.